

Introduction to Probability and Statistics

Lecture 6 Solutions

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Exercise 1: Adapted from Chapter 8, problem 25. It is claimed that a certain type of bipolar transistor has a mean value of current gain that is at least 210. A sample of these transistors is tested. If the sample mean value of the current gain is 200 and **the standard deviation is unknown, but estimated from the sample standard deviation** to be 35, would the claim be rejected at the 5 percent level of significance if

- a) The sample size is 25?
- b) The sample size is 64?

Note that the standard deviation is now unknown. How does this change the problem when compared to the one from lecture 4?

Solution:

- a) The sample size is 25?

The null hypothesis is $\mu \geq 210$ and we have that the sample mean is $\bar{X}_n = \hat{\mu}_n = 200$ and the sample standard deviation is $S_n = 35$.

Since we have only the estimated value of the standard deviation $\hat{\sigma}_n = S_n$, we need to perform a one-sided test using the t-distribution.

The test statistic for this type of one-sided test is $T = (\hat{\mu}_n - \mu_0)\sqrt{n}/S_n$ and the rejection region is $R = \{x : T(x) < c\}$. We also need to calculate the value of $c = -T_{n-1, \alpha}$.

Option 1: We get that $T = -1.4285$ and $c = -1.7108$. Since $T > c$, we **cannot reject** H_0 .

Option 2: We calculate the p-value as $v = P(T_{25-1} \leq T) = 0.083$. Since $v > \alpha$, we **cannot reject** H_0 .

b) The sample size is 64?

Now we have that $n = 64$ and re-do the calculations.

Option 1: We get that $T = -2.2857$ and $c = -1.6694$. Since $T < c$, we **reject** H_0 .

Option 2: We calculate the p-value as $v = P(T_{64-1} \leq T) = 0.0128$. Since $v < \alpha$, we **reject** H_0 .

Exercise 2: Chapter 8, problem 43

A question of medical importance is whether jogging leads to a reduction in one's heart rate. To test this hypothesis, 8 nonjogging volunteers agreed to begin a 1-month jogging program. After the month, their pulse rates were determined and compared with their earlier values. If the data are as follows, can we conclude that jogging has an effect on the pulse rates with a 5 percent level of significance?

Subject	1	2	3	4	5	6	7	8
Pulse rate before	74	86	98	102	78	84	79	70
Pulse rate after	70	85	90	110	71	80	69	74

Tips: Which type of test should you use? The paired t-test. How should you formulate the null-hypothesis so that it represents that jogging has no effect on pulse rates?

Solution:

The test to use is a paired two-sided t-test where the null hypothesis is $H_0 : \mu_A = \mu_B$.

We calculate the mean of the pulse rate before $\bar{X}_B = 81.125$ and after $\bar{X}_A = 83.875$ jogging. Then, we calculate the variance $S_{\bar{X}_A - \bar{X}_B}^2 = 159.554$.

With these values, we calculate the test statistic as

$$T = \frac{|\bar{X}_A - \bar{X}_B|}{S_{\bar{X}_A - \bar{X}_B} \sqrt{\frac{1}{n_A} + \frac{1}{n_B}}} = 0.435 \quad (1)$$

and the threshold is $c = T_{n_A+n_B-2, \alpha/2} = 2.145$. Since $T < c$, we **cannot reject the null hypothesis** H_0 that jogging has no effect on the pulse rate. Alternatively, we could calculate the

p-value

$$\begin{aligned}v &= \Pr(T_{n_A+n_B-2} \leq -T) + \Pr(T_{n_A+n_B-2} \geq T) \\&= 2(1 - \Pr(T_{n_A+n_B-2} \leq T)) \\&= 0.670,\end{aligned}$$

which is far greater than α .

Exercise 3:

Two different types of cable insulation have been tested. We need to determine the voltage level at which failures occur. The tests revealed that the individual cables failed at the following voltage levels.

TABLE I
VOLTAGE LEVEL AT WHICH FAILURES OCCURRED

Type A cables		Type B cables	
36	54	52	60
44	52	64	44
41	37	38	48
53	51	68	46
38	44	66	70
36	35	52	62
34	44		

We know that the voltage level that cables of type A can withstand is normally distributed with unknown mean μ_A and known variance $\sigma_A^2 = 40$. We also know that the voltage level that cables of type B can withstand is normally distributed with unknown mean μ_B and known variance $\sigma_B^2 = 100$.

With these data, can you reject the hypothesis $\mu_A = \mu_B$ with

- a) 5 percent level of significance?
- b) 10 percent level of significance?
- c) 1 percent level of significance?

Tips: Which type of test should you use? What happens to the result of a test when we increase or decrease the level of significance?

Solution:

Again, our null hypothesis is $H_0 : \mu_A = \mu_B$.

Besides, now we are given the variances $\sigma_A^2 = 40$ and $\sigma_B^2 = 100$ and we know that the values are normally distributed. Therefore, we can solve this problem with a z-test.

We know that the distribution of the difference of means with normal RVs and known variance is $N(\mu_A - \mu_B, \sigma_A^2/n_A + \sigma_B^2/n_B)$. Therefore, the difference of means has a distribution $N(-13.0476, 11.1904)$.

The test statistic is $T = |\mu_A - \mu_B| / \sqrt{\sigma_A^2/n_A + \sigma_B^2/n_B} = 3.9004$.

- a) Since $T = 3.9004 > Z_{\alpha/2}$ for $\alpha = 0.05$, the null hypothesis is rejected.
- b) Given that we rejected the null hypothesis with $\alpha = 0.05$, we can immediately conclude that we will reject the null hypothesis for all $\alpha' > \alpha = 0.05$, including the case where $\alpha' = 0.1$. You can make the calculations to confirm this.
- c) Even though the significance level is dropped to $\alpha = 0.01$, the threshold $c = Z_{0.005} = 2.575$ is still below the test statistic $T = 3.9004$. So, we also reject H_0 in this case.

Exercise 4: Chapter 9, problem 12

The following data set presents the heights of 12 male law school classmates whose law school examination scores were roughly equal. It also gives their annual salaries 5 years after graduation. Each of them went into corporate law. The height is in inches and the salary in units of \$1,000.

Height	Salary
64	91
65	94
66	88
67	103
69	77
70	96
72	105
72	88
74	122
74	102
75	90
76	114

- a) Do the above data establish the hypothesis that a lawyer's salary is related to his height?
Use the 5 percent level of significance.
- b) What was the null hypothesis in part a)

Solution:

- a) By performing linear regression on the data, we obtain the regression coefficients $\hat{\beta}_0 = -4.9857$ and $\hat{\beta}_1 = 1.4571$ and a coefficient of determination $R^2 = 0.2341$. Therefore, the data exhibits a linear relationship with positive slope. That is, a linear and positive correlation.
- b) To make this test, we start with the null hypothesis being $H_0: \beta_1 = 0$. If H_0 is rejected, we can confidently say that the data does not support the claim of a linear relationship. To proceed with the test, we find the standard error $\hat{s}_e = 0.8334$, our test statistic $T = \left| \hat{\beta}_1 - 0 \right| / \hat{s}_e = 1.7483$, and the t-value $T_{n-2, \alpha/2}$. By setting $\alpha = 0.05$, we obtain $T_{n-2, \alpha/2} = 2.2281$. In a two-sided t-test, the rejection region is $T > T_{n-2, \alpha/2}$. Since this is not the case, we cannot reject the null hypothesis. That is, the evidence is not sufficient to reject the claim, even when there is no real linear relationship between the height and the salary.